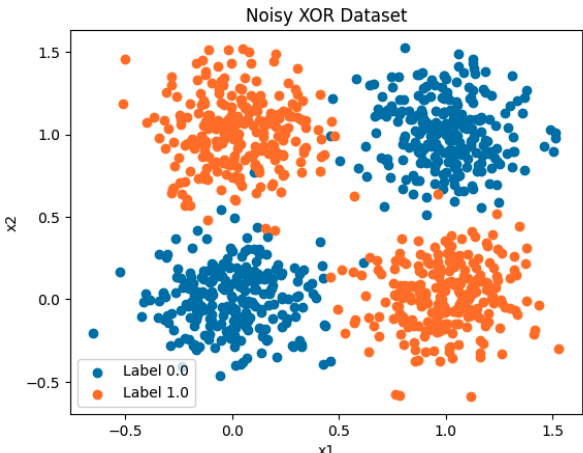




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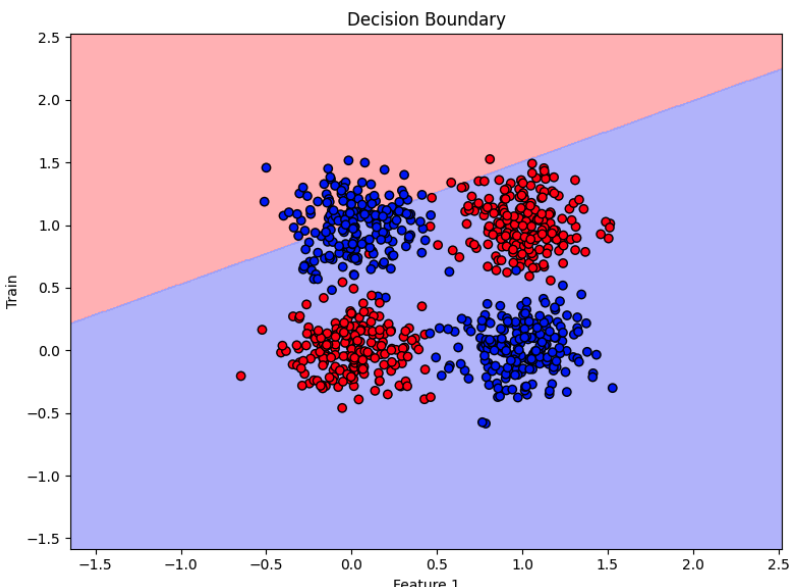
2 See codes

3 0.4

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5 The boundary doesn't fit the data well. This is obvious because we can see the dataset consists of

6 perceptron2 = Sequential()perceptron2.add(Dense(units=8, activation='relu', input_dim=2, kernel_in

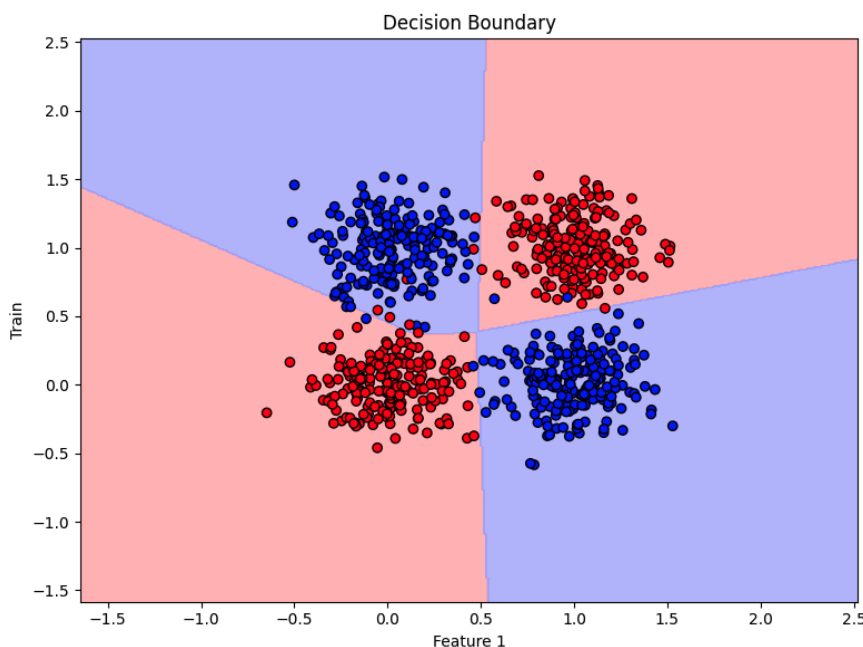
7 0.99

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9 The number of neurons in the hidden layer is a key design choice that affects a neural network's

10 hidden_dim test_accuracy0 2 0.9101 4 0.9802 6 0.9803

11 The best hidden layer dimension is 12.0, with a validation accuracy of 0.98.

12 Model Results: Model Architecture Validation Accuracy0 [2, 4, 1] 0.8751 [2, 4, 4

13 Final Training Accuracy: 100.0000%Final Test Accuracy: 100.0000%Best Validation Accuracy durin

14 Training Accuracy: 1.0000Validation Accuracy: 1.0000

15 A high learning rate can cause multiple issues in gradient descent. In full-batch gradient

16 Random horizontal flipping is a data augmentation method that is suitable for classifying images of

17 The advantage of Network A lies in its simpler architecture, consisting of only two fully connected

18 See codes

19 See codes

20 See codes

21 Accuracy: 0.8192 Precision: 0.6654 Recall: 0.3645 F1 Score: 0.4710

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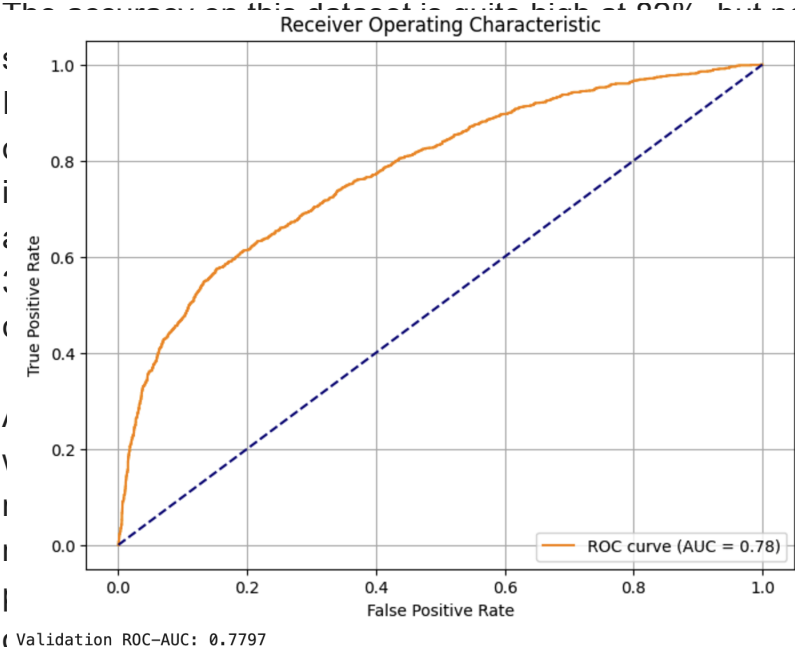
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Recall will hence focus on reducing false negatives. The low value of recall, 36.45%, shows that a model fails to identify about two-thirds of the actually defaulted cases. F1 Score is the harmonic mean of Precision and Recall. It offers a balanced measure of both Precision and Recall, which is better than the case when any one of them is used as a trade-off against another. The F1 score of 47.10% in this data represents a balance between precision and recall, showing decent precision but a recall that's in need of some considerable improvement. ROC-AUC: It is a measure of the model's capability in distinguishing between positive and negative classes at any threshold level. In this problem, an area under the ROC curve or AUC of 0.7797 for the separation between

22 default and non-default cases is obtained, which is good. Unlike precision and recall, ROC-AUC is

23 See codes

24 Accuracy is the ratio of correctly classified instances (positive and negative) to total instances,

25 See codes

26 Recall and F1 score are of prime importance considering the nature of data and the very essence

27 Validation Metrics: Accuracy: 0.8121, Precision: 0.6070, Recall: 0.4220, F1 Score: 0.4979

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Previous Assignment Metrics: Accuracy: 0.7832, Precision: 0.5101, Recall: 0.5819, F1 Score: 0.5436
Current Model Metrics: Accuracy: 0.8121, Precision: 0.6070, Recall: 0.4220, F1 Score: 0.4979. The previous assignment metrix is based on logistics regression. Neural Network achieves better accuracy (81.68%) compared to previous (78.32%). Precision has also improved in the Neural Network (62.36%) compared to previous (51.01%).Recall is lower for the Neural Network (42.99%) compared to previous (58.19%).The F1 Score for the Neural Network (50.89%) is slightly lower than previous (54.36%). The Neural Network is better at identifying true positives without mistakenly labeling negatives as positives and it is better at identifying true positives without mistakenly labeling negatives as positives. Also the Neural Network is missing more true positives compared to previous. Since F1 Score balances precision and

28 recall, the Neural Network's lower recall significantly impacts this metric despite its improved precision.

29 See codes

30 See codes

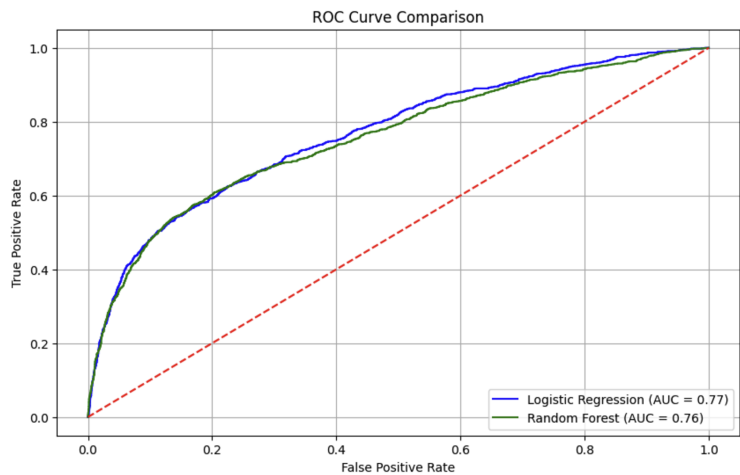
31 Accuracy: 0.8190, Precision: 0.6707, Recall: 0.3544, f1_score: 0.4638

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33 Random Forest, meaning it is better at identifying defaulters. And it has highest F1 score (0.4979), which

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